

Chapter Five: Market Structure

- ❑ This chapter discusses how a particular firm makes a decision to achieve its profit maximization objective.
- ✓ A firm's decision to achieve this goal is dependent on the type of market in which it operates.
- ❑ To this effect we distinguish between four major types of markets:
 - ✓ perfectly competitive market,
 - ✓ monopolistically competitive market,
 - ✓ oligopolistic market, and
 - ✓ pure monopoly market.

5.1. The concept of market in physical and digital space

- ✓ Comprehensive definition of market according to American Marketing Association (1985) is the process of planning and executing the conception, pricing, promotion, and distribution of goods, services and ideas to create exchanges.

❑ So market describes **place** or **digital space** by which goods, services and ideas are exchanged to satisfy consumer need.

Digital marketing is the marketing of products or services using **digital technologies**, mainly on the **internet** but also including **mobile phones**, **display advertising**, and **any other digital media**.

✓ Digital marketing channels are systems on the internet that can create, accelerate and transmit product value from producer to the terminal consumer by digital networks.

Physical market is a set up where buyers can physically meet their sellers and purchase the desired merchandise from them in exchange of money.

✓ In physical marketing, marketers will effortlessly reach their target local customers and thus they have more personal approach to show about their brands.

✓ The **choice of the marketing** mainly depends **on the nature of the products and services**.

- ❑ In simple words, we may define market as “a structure in which the buyers and sellers of a commodity remain in contact.”
- ❖ Markets are classified into different types on the basis of factors such as:
 - ✓ the **degree of competition** among firms in a market,
 - ✓ the **number of buyers and sellers**, the nature of the commodity,
 - ✓ the **mobility of goods and factors of production**, and
 - ✓ the knowledge of buyers and sellers about prices in the market(**Information**)

5.2. Perfectly competitive market

- ✓ **Perfect competition** is a market structure in which there are a large number of producers (firms) producing a homogeneous product so that no individual firm can influence the price of the commodity.
- ✓ In this type of market, the **price is determined by the industry** (aggregate of all the firms producing the same product) through the forces of demand and supply

5.2.1 Assumptions of perfectly competitive market

A market is said to be pure competition (perfectly competitive market) if the following assumptions are satisfied.

1. Large number of sellers and buyers:

- ✓ under perfect competition the number of sellers is assumed to be too large that the share of each seller in the total supply of a product is very small.
- ✓ Therefore, no single seller can influence the market price by changing the quantity supply.
- ✓ Similarly, the number of buyers is so large that the share of each buyer in the total demand is very small and that no single buyer or a group of buyers can influence the market price by changing their individual or group demand for a product.
- ✓ Therefore, in such a market structure, sellers and buyers are not price makers rather they are price takers, i.e., the price is determined by the interaction of the market supply and demand forces.

2. Homogeneous product

- ✓ Products sold in the market are homogeneous – i.e., they are **identical** in all respects, including **quality**, **colour**, **size**, **weight**, **design**, etc.
- ✓ Buyers perceive **no actual** or **real differences between the products** offered by different firms.
- ✓ Product of each firm is regarded as **a perfect substitute** for the products of other firms.
- ✓ Therefore, **no firm can gain any competitive advantage** over the other firm.

Example: Chemical inputs

3. Free entry and exit

- ✓ There is **no restriction** or **market barrier** on entry of new firms to the industry, and no restriction on exit of firms from the industry.
- ✓ A firm may enter the industry or quit it **on its accord**.

3. Perfect mobility

- ✓ factors of production are free to move from one firm to another throughout the economy.
- ✓ No **hindrance** or **obstruction**.
- ✓ This means that **labour can move from one job to another** and from one region to another.
- ✓ Capital, raw materials, and other factors are not monopolized.

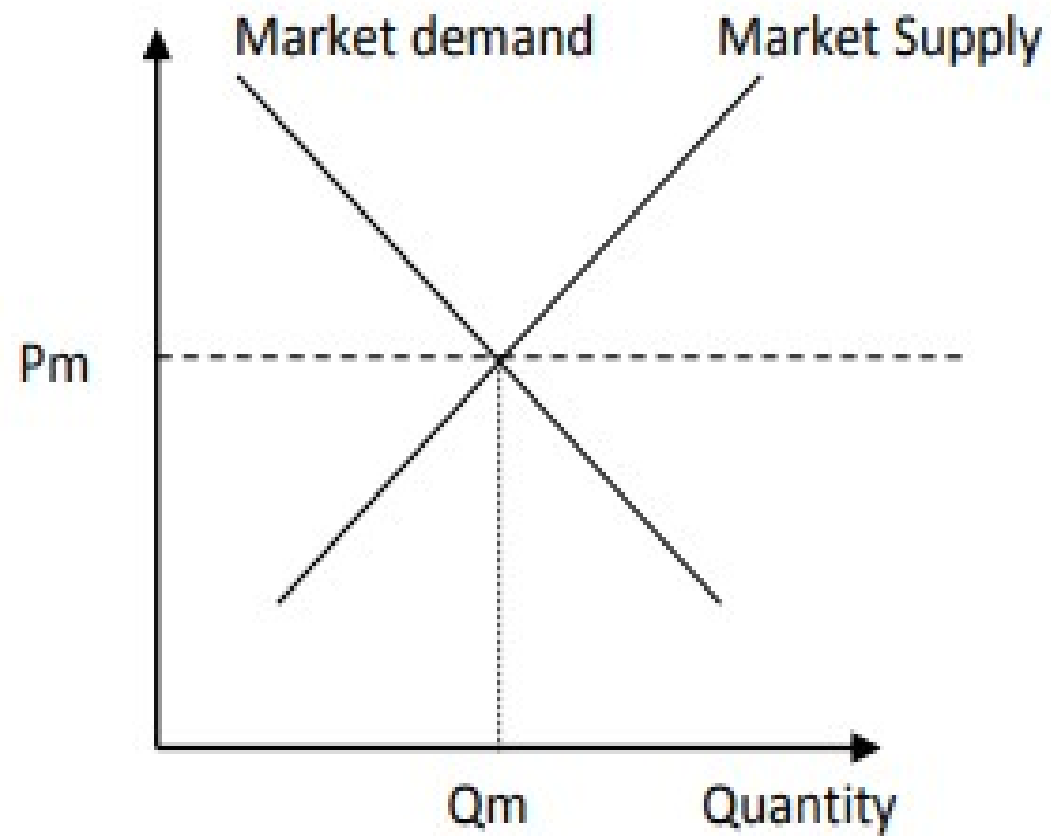
5. Perfect knowledge about market conditions:

- ✓ all the buyers and sellers have **full information** regarding the prevailing and future prices and availability of the commodity.

6. No government interference:

- ✓ Government does not interfere in any way with the **functioning of the market**.
- ✓ There are **no discriminator taxes** or **subsidies**, **no allocation of inputs** by the procurement, or any kind of direct or indirect control.

- ✓ That is, the government follows the free enterprise policy.
- ✓ Where there is intervention by the government, it is intended to correct the market imperfection.
- ✓ From these assumptions, **a single producer(firm)** under perfectly competitive market is **a price-taker**.
- ✓ That is, at the market price, **the firm can supply whatever quantity it would like to sell**.
- ✓ Once the price of the product is determined in the market, the producer takes the price (P_m in the **figure 5.1**) as given.
- ✓ Hence, the demand curve **D_f** that the firm faces in this market situation is **a horizontal line** drawn at the equilibrium price, **P_m**



(a): Market demand and supply



(b): Demand curve of a firm

Figure 5.1: Individual and market demand curve

- ❑ The **main objective** of a firm is **profit maximization**.
 - ✓ If the firm has to incur a loss, it aims to minimize the loss.
 - ✓ **Profit** is the difference between total revenue and total cost.
- ❖ **Total Revenue (TR)**: it is the total amount of money a firm receives from a given quantity of its product sold.
 - ✓ It is obtained by multiplying the unit price of the commodity and the quantity of that product sold.

$$TR = P \times Q$$

where P = price of the product Q = quantity of the product sold.

- ❖ **Average Revenue (AR)**: it is the revenue per unit of item sold.
 - ✓ It is calculated by dividing the total revenue by the amount of the product sold.

$$AR = \frac{TR}{Q} = \frac{P \cdot Q}{Q}; AR = P$$

- ✓ Therefore, the firm's demand curve is also **the average revenue curve**

- ❖ **Marginal Revenue:** it is the additional amount of money/ revenue the firm receives by selling one more unit of the product.
- ✓ In other words, it is the change in total revenue resulting from the sale of an extra unit of the product.
- ✓ It is calculated as the ratio of the change in total revenue to the change in the sale of the product.

$$\mathbf{MR} = \frac{\Delta TR}{\Delta Q} = \frac{\Delta P \cdot Q}{\Delta Q} = \frac{P \Delta Q}{\Delta Q} = \mathbf{P}$$

- Thus, in **a perfectly competitive market**, a firm's **average revenue, marginal revenue** and **price of the product** are equal.

$$\text{i.e. } \mathbf{AR} = \mathbf{MR} = \mathbf{P} = \mathbf{D}_f$$

- Since the **purely competitive firm** is **a price taker**, it will maximize its economic profit **only by adjusting its output.**
- In the short run, the firm has **a fixed plant.**

- ✓ Thus, it can adjust its output only through changes in the amount of **variable resources**.
- ✓ It adjusts its variable resources to achieve the output level that maximizes its profit.
- There are **two ways** to determine the level of output at which a competitive firm will realize **maximum profit** or **minimum loss**.
 - ✓ One method is to compare **total revenue** and **total cost**
 - ✓ the other is to compare **marginal revenue** and **marginal cost**.

A) Total Approach (TR-TC approach)

- ✓ In this approach, a firm maximizes total profits in the short run **when the (positive) difference** between total revenue (TR) and total costs (TC) is **greatest**.

$$\text{Total Profit} = \text{Total Revenue} - \text{Total Cost}$$

$$TP = TR - TC$$

Note: The profit maximizing output level is Q_e because it is at this output level that the vertical distance between the **TR** and **TC** curves (or profit) is maximized.

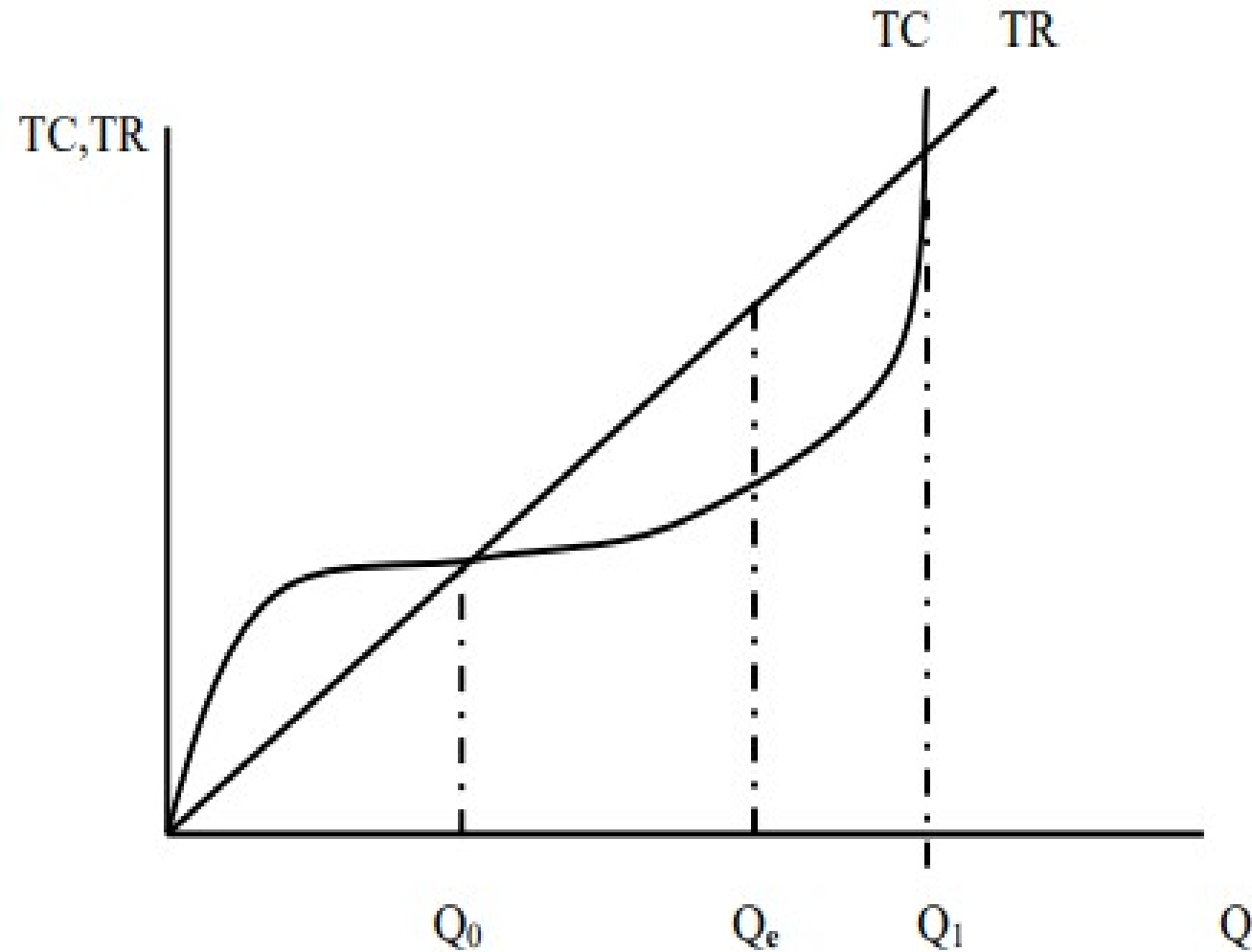


Figure 5.2: Total revenue and total cost approach of profit maximization

B) Marginal Approach (MR-MC)

- ✓ In the short run, the firm will maximize profit or minimize loss by producing the output at which marginal revenue equals marginal cost.
- ✓ More specifically, the **perfectly competitive firm maximizes** its at the output when the following two conditions are met:
- ✓ **short-run total profits**
 - i) **$MR = MC$**
 - ii) The **slope of MC** is greater than **slope of MR**; or **MC is rising** (that is, slope of MC is greater than zero).

✓ Mathematically, $\Pi = TR - TC$

Π is maximized when $\frac{d\Pi}{dQ} = 0$

That is $\frac{d\Pi}{dQ} = \frac{dTR}{dQ} - \frac{dTC}{dQ} = 0, MR - MC = 0;$

$MR = MC \dots \dots \dots (First\ Order\ Condition(FOC))$

Then,

$$\frac{d^2\Pi}{d^2Q} < 0; \frac{d^2\Pi}{d^2Q} = \frac{d^2TR}{d^2Q} - \frac{d^2TC}{d^2Q} < 0,$$

$$\frac{d^2\Pi}{d^2Q} < \frac{dMR}{dQ} - \frac{dMC}{dQ} < 0$$

$$\frac{dMR}{dQ} = \text{Slope of } MR; \text{ and } \frac{dMC}{dQ} = \text{Slope of } MC$$

□ Therefore, **Slope of MC > slope of MR** ---...**Second order condition (SOC)**

□ Slope of **MC > 0** (because the **slope of MR is zero**)

Graphically, the marginal approach can be shown as follows.

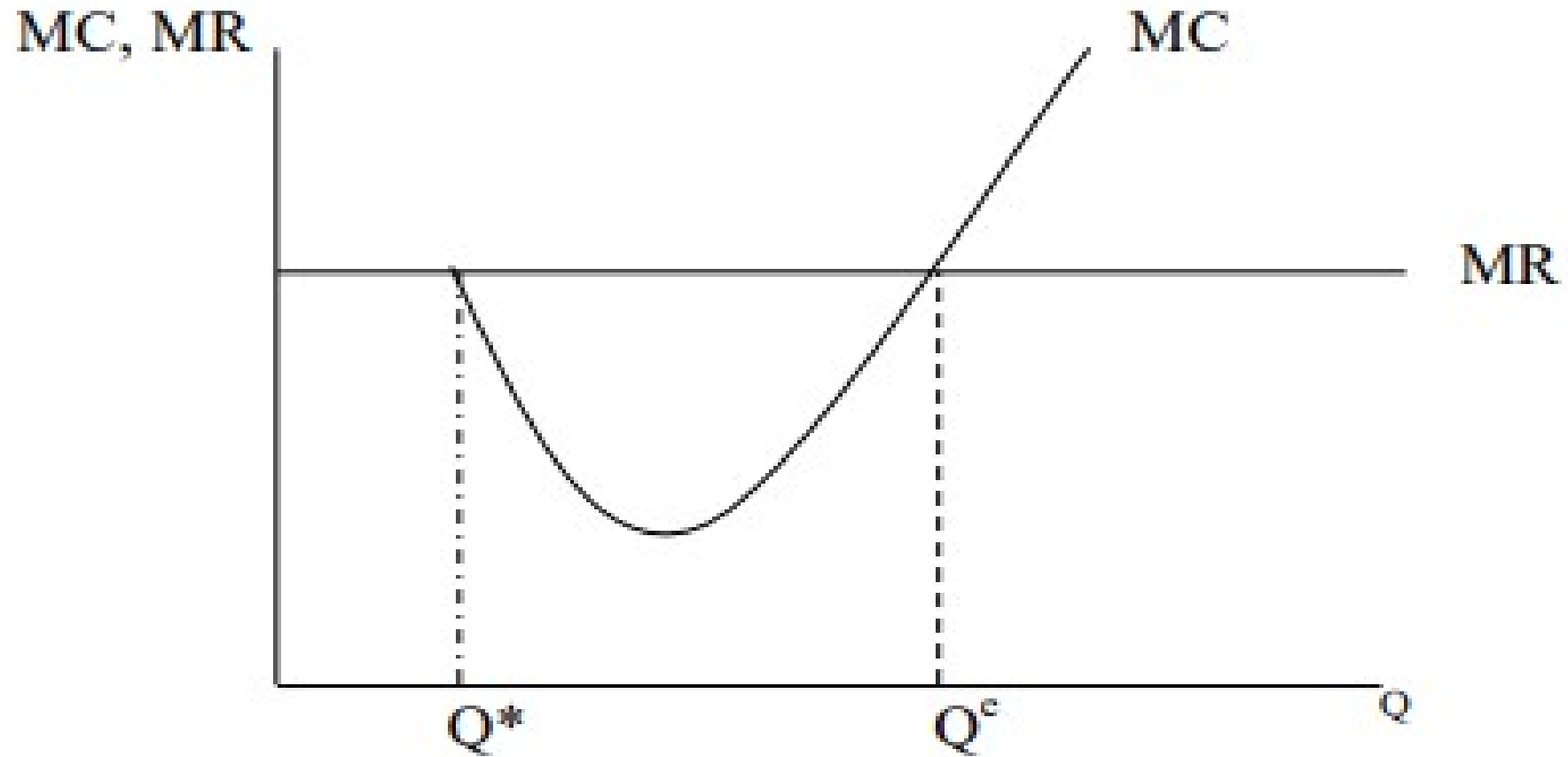


Figure 5.3: *Marginal approach of profit maximization*

- ✓ The profit maximizing output is Q^e , where $MC = MR$ and **MC curve is increasing**.
- ✓ At Q^* , $MC = MR$, but since **MC is falling** at this output level, it is not equilibrium output.
- Whether the firm in the short-run gets positive or zero or negative profit depends **on the level of ATC at equilibrium**.
- ✓ Thus, depending on the relationship between price and ATC, the firm in the short-run may earn **economic profit**, **normal profit** or **incur loss** and decide to **shut-down business**.

i) **Economic/positive profit**

- ✓ If the **AC** is below the **market price at equilibrium**, the firm earns a positive profit equal to **the area between the ATC curve and the price line up to the profit maximizing output**.

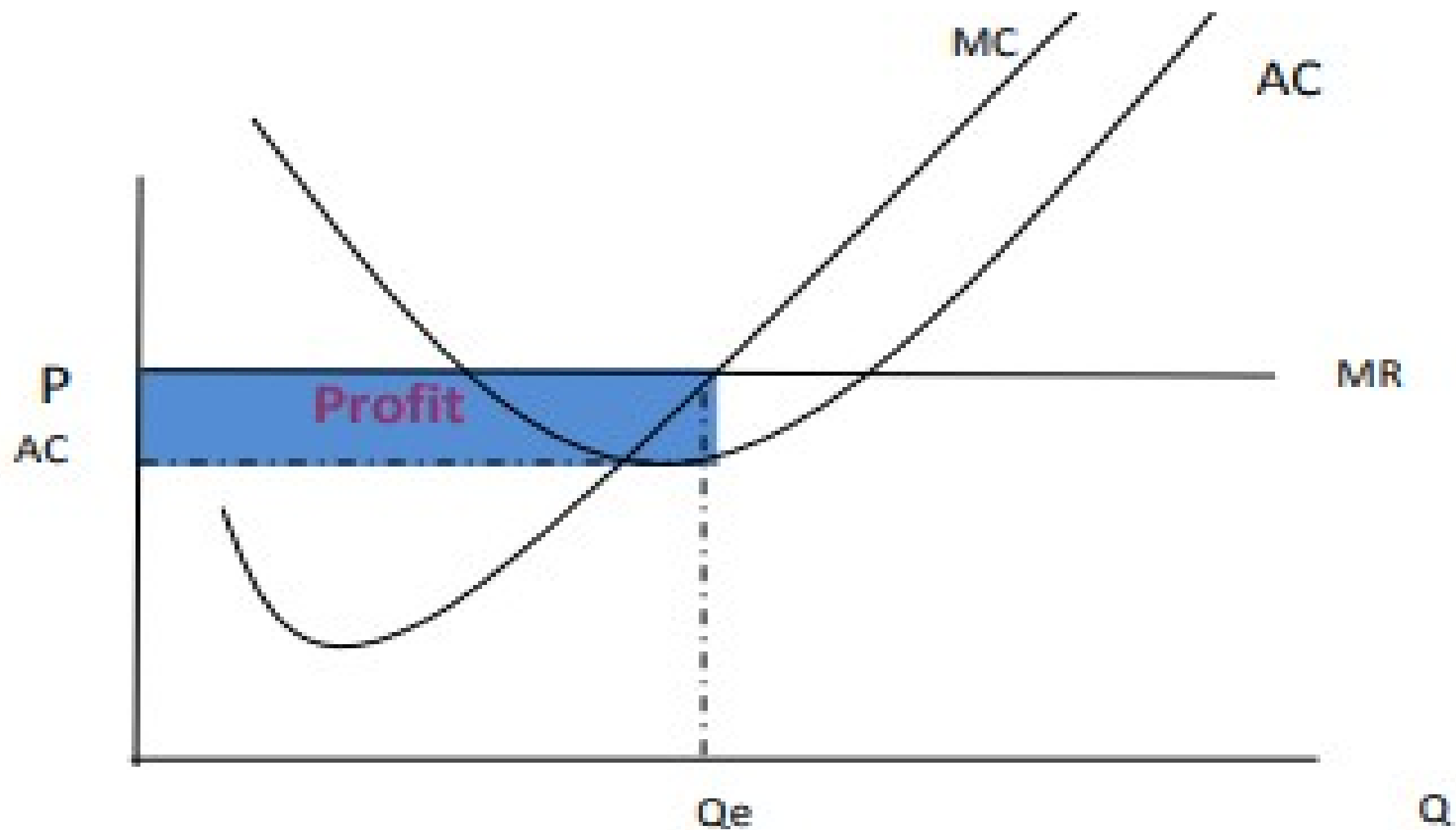


Figure 5.4: Economic profit a firm

ii) Loss

- ✓ If the **AC** is above the market price at equilibrium, the firm earns a **negative profit** (incurs a loss) equal to the area between the **AC curve** and the price line.

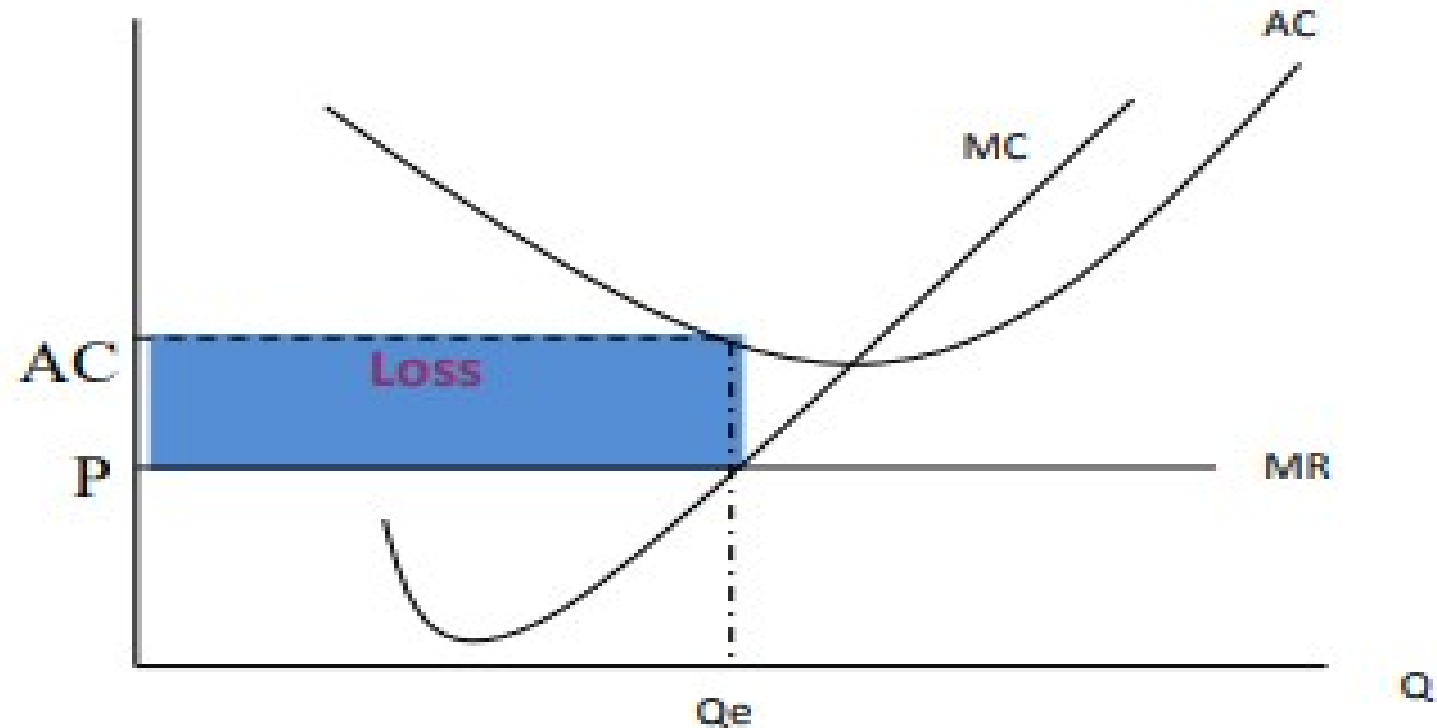


Figure 5.6: A firm incurring a loss

iii) Normal Profit (zero profit) or break- even point

- ✓ If the **AC** is equal to the **market price at equilibrium**, the firm gets zero profit or normal profit.

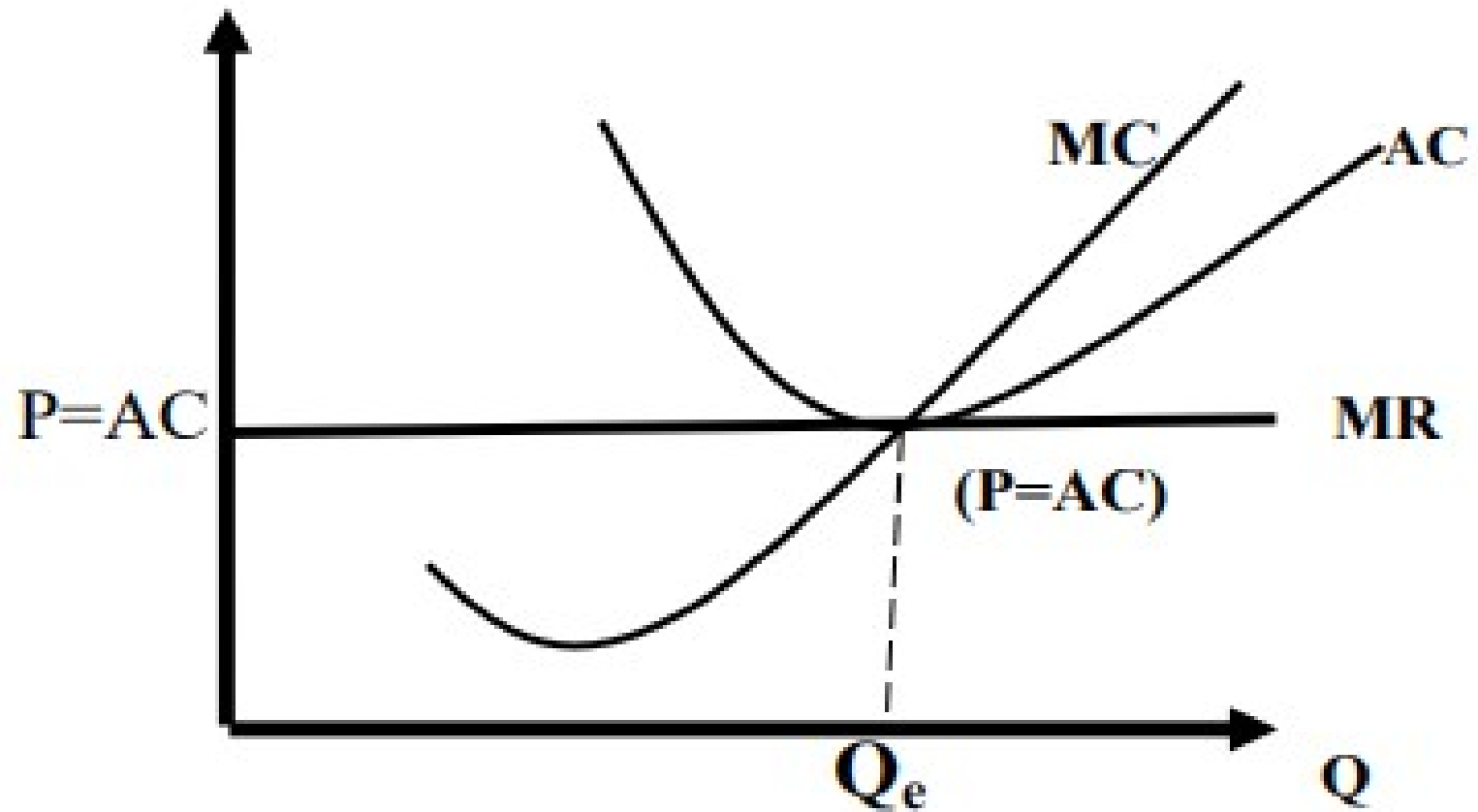


Figure 5.7: A firm earning a normal profit

• iv) Shutdown Point

- ✓ The firm will not stop production simply because AC exceeds price in the short-run.
- ✓ The firm will continue to produce irrespective of the existing loss **as far as the price is sufficient to cover the average variable costs (AVC)**.
- ✓ This means, if P is larger than AVC but smaller than AC , **the firm minimizes total losses**.
- ✓ But if P is smaller than AVC , the firm minimizes total losses by shutting down.
- ✓ Thus, **$P = AVC$** is the **shutdown point for the firm**.

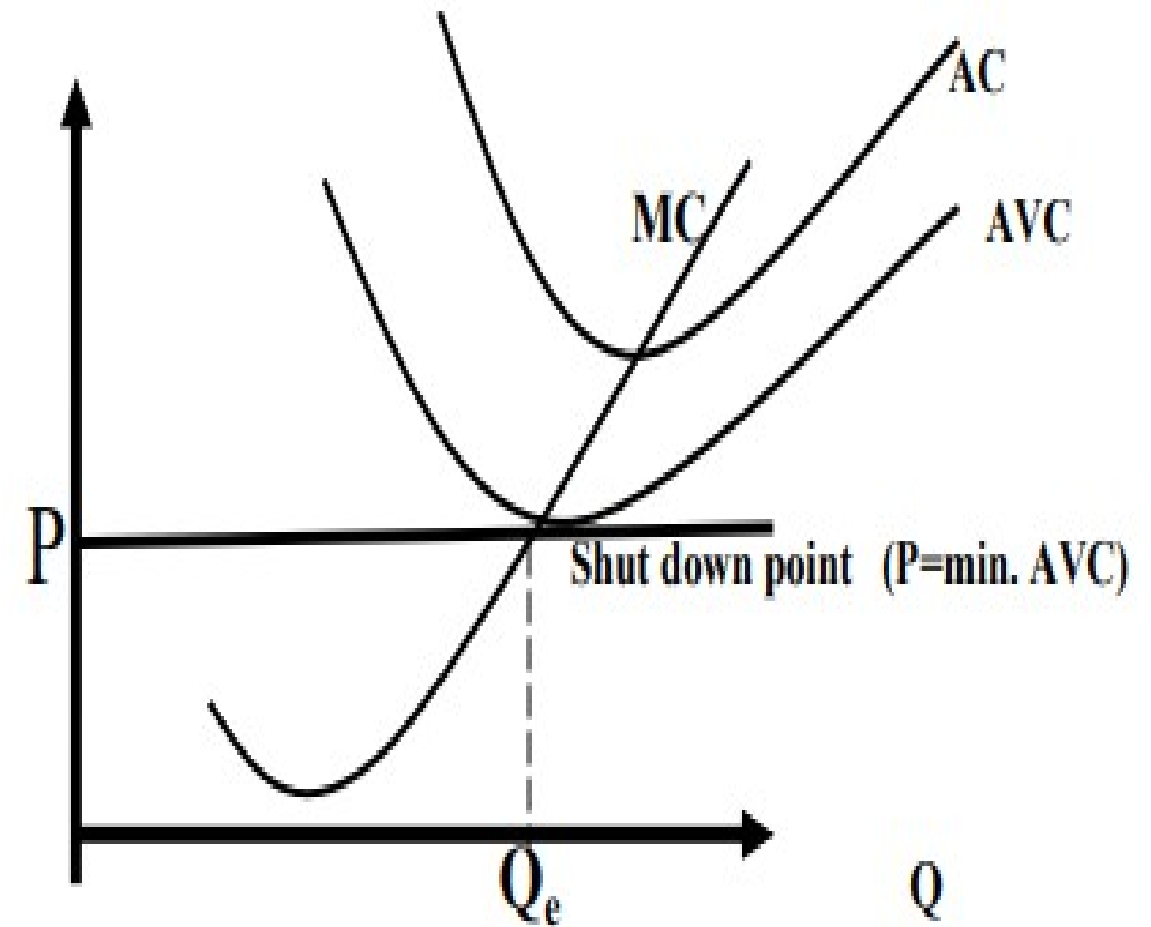


Figure 5.8: A shut down point

Example: Suppose that the firm operates in **a perfectly competitive market**. The market price of its product is \$10. The firm estimates its cost of production with the following cost function: **$TC = 2 + 10Q - 4Q^2 + Q^3$**

- A) What level of output should the firm produce to maximize its profit?
- B) Determine the level of **profit at equilibrium**.
- C) What **minimum price** is required by the firm to stay in the market?

Solution

Given: **$P = \$10$** and **$TC = 2 + 10Q - 4Q^2 + Q^3$**

A) The profit maximizing output is that level of output which satisfies the following condition:

- $i) MC = MR$ &**
- $ii) MC$ is rising**

- ✓ Thus, we have to find **MC** & **MR** first.
- ✓ MR in a perfectly competitive market is equal to the market price.
- ✓ Hence, **MR=10**

❖ **Alternatively, MR** $= \frac{dTR}{dQ} = \frac{dP*Q}{dQ} = \frac{d(10Q)}{dQ} = 10$

❖ **MC** $= \frac{dTC}{dQ} = \frac{d(\mathbf{2+10Q-4Q^2+Q^3})}{dQ} = \mathbf{10 - 8Q + 3Q^2}$

- ✓ **At equilibrium, MC = MR**

$$10 - 8Q + 3Q^2 = 10$$

$$Q(-8 + 3Q) = 0$$

$$\mathbf{3Q = 8 \text{ and } Q = 0}$$

$$\mathbf{Q = \frac{8}{3}; Q = 0}$$

- ✓ Now we have obtained **two different output levels** which satisfy the **first order (necessary) condition** of profit maximization.

- ✓ To determine which level of output maximizes profit we have to use the **second order test** at the two output levels.
- ✓ That is, we have to see which output level satisfies the second order condition of **increasing MC**.
- ❖ To see this first we determine the **slope of MC**.

$$\text{Slope of MC} = \frac{dMC}{dQ} = -8 + 6Q$$

- ✓ At **$q = 0$** , slope of MC is $-8 + 6(0) = -8$ which is **negative**, implying that marginal cost is decreasing at $q = 0$.
- ✓ Thus, **$q = 0$** is not equilibrium output because it doesn't satisfy the second order condition.
- ✓ At **$q = 8/3$** , slope of MC is $-8 + 6(8/3) = 8$, which is **positive**, implying that MC is increasing at **$q = 8/3$**
- ✓ **Thus, the equilibrium output level is $q = 8/3$**

B) Above, we have said that the firm maximizes its profit by producing $8/3$ units.

- ✓ To determine the firm's equilibrium profit we have to calculate the **total revenue** that the firm obtains at this level of output and the **total cost** of producing the equilibrium level of output.

$$\mathbf{TR} = \text{Price} * \text{Equilibrium Output} = \$ 10 * 8/3 = \mathbf{\$ 80/3}$$

- ✓ TC at $q = 8/3$ can be obtained by substituting $8/3$ for q in the TC function, i.e.,

$$\mathbf{TC = 2 + 10\left(\frac{8}{3}\right) - 4(8/3)^2 + \left(\frac{8}{3}\right)^3 = 19.18}$$

- ✓ Thus the equilibrium (maximum) profit is

$$\begin{aligned}\mathbf{\Pi} &= \mathbf{TR - TC} \\ \mathbf{\Pi} &= \mathbf{26.67 - 19.18 = \$ 7.48}\end{aligned}$$

C) To stay in operation the firm needs the price which equals **at least the minimum AVC**.

- ✓ Thus, to determine the minimum price required to stay in business, we have to determine the **minimum AVC**.
- ✓ AVC is minimal when **derivative of AVC** is equal to zero.

$$\text{That is: } \frac{dAVC}{dQ} = 0$$

❖ Given the TC function: **$TC = 2 + 10Q - 4Q^2 + Q^3$** , **$TVC = 10Q - 4Q^2 + Q^3$** .

$$\checkmark \text{ Then, } AVC = \frac{TVC}{Q} = \frac{10Q - 4Q^2 + Q^3}{Q} = 10 - 4Q + Q^2$$

$$\checkmark \frac{dAVC}{dQ} = \frac{d(10 - 4Q + Q^2)}{dQ} = -4 + 2Q = 0$$

$$Q = 2$$

- ✓ Thus, **AVC** is minimum when output is equal to **2 units**.

5.2.3 Short run equilibrium of the industry

- ✓ Since the **perfectly competitive firm** always produces where $P = MR = MC$ (as long as P exceeds AVC), the **firm's short-run supply curve** is given by the **rising portion of its MC curve above its AVC**, or shutdown point (see figure 5.7).
- ✓ The **industry/market supply curve** is a **horizontal summation** of the **supply curves of the individual firms**.
- ✓ Industry supply curve can be obtained by multiplying the individual supply at various prices by the number of firms, if firms have **identical supply curve**.
- ✓ An industry is in equilibrium in the short-run when market is cleared at a given price i.e. when the total **supply = total demand** for its product, the prices at which market is cleared is **equilibrium price**.
- ✓ When an industry reaches **at its equilibrium**, there is **no tendency to expand or to contract the output**.

5.3. Monopoly market

5.3.1. Definition and characteristics

- ✓ This is at the opposite end of the spectrum of market structures.
- ✓ Pure monopoly exists when **a single firm** is **the only producer of a product** for which there are no close substitutes.
- The main characteristics of this market structure include:

1. Single seller

- ✓ A pure or absolute monopoly is a one firm industry
- ✓ A single firm is the only producer of a specific product or the **sole supplier of the product**, and
- ✓ The **firm** and the **industry** are synonymous.

2. No close substitutes

- ✓ the monopolist's **product is unique** in that there are **no good or close substitutes**.
- ✓ From the buyer's view point, there are no reasonable alternatives.

3. Price maker

- ✓ the individual firm exercises a considerable **control over price** because it is responsible for, and therefore controls, the total quantity supplied.
- ✓ Confronted with the usual **down ward sloping demand curve** for its product, the monopolist **can change product price** by changing the quantity of the product supplied.

4. Blocked entry

- ✓ A pure monopolist has **no immediate competitors** because **there are barriers**, which keep potential competitors from entering in to the industry.
- ✓ These barriers may be **economic**, **legal**, **technological** etc.
- ✓ Under conditions of **pure monopoly**, **entry is totally blocked**.

5.3.2. Sources of monopoly

- ✓ The emergence and survival of monopoly is attributed to the factors which prevent the entry of other firms in to the industry.
- ✓ The **barriers** to entry are therefore the **sources of monopoly** power.
- The major sources of barriers to entry are:
 - i) **Legal restriction**
 - ✓ Some monopolies are created by **law** in **public interest**.
 - ✓ Such monopoly may be created in both public and private sectors.
 - ✓ Most of the state monopolies in the public utility sector, including postal service, **Natural gas companies**, telegraph, telephone services, **radio** and **TV** services, **generation** and **distribution of electricity**, **rail ways**, **airlines** etc... are **public monopolies**.

ii) Control over key raw materials

- ✓ Some firms acquire monopoly power from their traditional control over certain **scarce and key raw materials** that are essential for the production of certain other goods.
- ✓ **For example**, **Aluminum Company of America** had monopolized the **aluminum industry** because it had acquired control over almost all sources of bauxite supply; such monopolies are often called **raw material monopolies**.

iii) Efficiency

- ✓ a primary and technical reason for growth of monopolies is **economies of scale**.
- ✓ The most efficient plant (**probably large size firm**), which produces at minimum cost, can eliminate the competitors by curbing down its price for a short period and can acquire monopoly power.
- ✓ Monopolies created through efficiency are known as **natural monopolies**.

iv) Patent rights

- ✓ Patent rights are granted by the government to a firm to produce **commodity of specified quality** and **character** or to use specified rights to produce the specified commodity or to use the specified technique of production.
- ✓ Such monopolies are called to **patent monopolies.**

5.4. Monopolistically competitive market

- ✓ This market model can be defined as the market organization in which there are relatively **many firms selling differentiated products.**
- ✓ It is the blend of **competition** and **monopoly.**
- ✓ The competitive element arises from the existence of **large number of firms** and **no barrier to entry or exit.**

❑ This market is characterized by:

(i) Differentiated product

- ✓ the product produced and supplied by many sellers in the market is **similar but not identical in the eyes of the buyers.**
- ✓ **A variety of the same product.**
- ✓ The difference could be in **style**, **brand name**, in **quality**, or others.
- ✓ Hence, the **differentiation of the product could be real** (eg. quality) or fancied (e.g. difference in packing).

(ii) Many sellers and buyers

- ✓ there are many sellers and buyers of the product, but their number is not as large as that of the perfectly competitive market.

(iii) Easy entry and exit

- ✓ like the PCM, there is no barrier on new firms that are willing and able to produce and supply the product in the market.

- ✓ On the other hand, if any firm believes that it is not worth to stay in the business, it may exit.

(iv) Existence of non-price competition

- ✓ Economic rivals take the form of non-price competition in terms of product quality, advertisement, brand name, service to customers, etc.
- ✓ A firm spends money in advertisement to reach the consumers about the relatively unique character of its product and thereby get new buyers and develop brand loyalty.
- ✓ Many retail trade activities such as clothing, shoes, soap, etc are in this type of market structure.

5.5. Oligopoly market

- ✓ This is a market structure characterized by:

i) Few dominant firms

- ✓ there are few firms although the exact number of firms is undefined. Each firm produces a significant portion of the total output.

ii) Interdependence

- ✓ since few firms hold a significant share in the total output of the industry, each firm is affected by the price and output decisions of rival firms.
- ✓ Therefore, the distinguishing characteristic of oligopoly is the **interdependence among firms** in the industry.

iii) Entry barrier

- ✓ there are considerable obstacles that hinder a new firm from producing and supplying the product.
- ✓ The barriers may include **economies of scale**, **legal**, **control of strategic inputs**, etc.

iv) homogenous or differentiated Product

- ✓ If the product is homogeneous, we have **a pure oligopoly**.
- ✓ If the product is differentiated, it will be **a differentiated oligopoly**.

V) Lack of uniformity in the size of firms

- ✓ Firms differ considerably in size.
 - ✓ Some may be small, others very large.
 - ✓ Such a situation is **asymmetrical**.
 - ✓ **Non-price competition:** firms try to avoid price competition due to the fear of price wars and hence depend on non-price methods like advertising, after sales services, warranties, etc.
 - ✓ This ensures that firms can influence demand and build brand recognition.
- A special type of oligopoly in which there are **only two firms** in the market is known as **duopoly**.

Summarized Differences of the FOUR MARKETS

Characteristics	Market models			
	Pure Competition	Monopolistic Competition	Oligopoly	Pure Monopoly
Number of firms	Large	Many	Few	One or Single
Type of product	Homogeneous	Differentiated	Homogeneous or differentiated	Unique, no close substitutes
Control over price	None	Some, but within rather narrow limits	Limited by mutual interdependence and collusion	Significant
Condition of entry	Very easy	Relatively easy	Considerable barriers/obstacles	Blocked
Examples	Agricultural products	Clothes, Shoes	Steel, Automobiles	Local utilities